FACULTY OF ENGINEERINGAND NATURAL SCIENCES

CAPSTONE PROJECT PROPOSAL

AI- SUPPORTED SYLLABUS MANAGEMENT WEB PLATFORM

Gamze Demirci , Industrial Engineering
Gülmen İlayda Özyurt , Industrial Engineering
İrem Uluslu , Industrial Engineering
Zeynep İlgi Altunsoy, Software Engineering
Ece Eraslan, Software Engineering
Hatice Beril Satıcı, SoftwareEngineering

Advisors: Asst. Prof. Oğuzhan Erdinç,Asst. Prof. Özge Yücel Kasap

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STUDENT DECLARATION

By submitting this report, as partial fulfilment of the requirements of the Capstone course, the students promise on penalty of failure of the course that

- they have given credit to and declared (by citation), any work that is not their own (e.g. parts of the report that is copied/pasted from the Internet, design or construction performed by another person, etc.);
- they have not receivedunpermitted aid for the project design, construction, report or presentation.
- they have not falselyassigned credit for work to another student in the group and not take credit for work donebyanother student in the group.

DECLARATION OF AI USAGE

Tools Used During the preparation of this Capstone Project Proposal, I utilized the following tools: Chatgpt

Purpose and Methodology AI was used exclusively for: Ideation, language refinement, checking and correcting minor grammar errors and structure academic content. The tool supported the organization of ideas and improvement of clarity in early drafts.

Example: "The tool provided assistance with grammar correction and content organization."

Prompts used

Example: " Check this paragraph for grammar and spelling mistakes without changing the meaning"

Example: "Rearrange paragraphs to improve academic structure while keeping the original meaning."

Validation and Authorship I certify that:

I have personally verified all facts, citations, and data points provided by the AI.

I have not simply copied and pasted AI outputs; all final text is my own writing or a significant rewrite of AI suggestions.

I accept full responsibility for the integrity and accuracy of this proposal.

ProjectTeam

Department	***1***	***:Industrial Engineering***
Member	***1***	***:Gamze Demirci***
Member	***2***	***:Gülmen İlayda Özyurt***
Member	***3.***	***:İrem Uluslu***
Advisor	***cc***	***:Asst. Prof. Oğuzhan Erdinç***
Department2		***: Software Engineering***
Member	***1***	***:Zeynep İlgi Altunsoy***
Member	***2***	***:Hatice Beril Satıcı***
Member	***3***	***:Ece Eraslan***
Advisor		***:Asst. Prof. Özge Yücel Kasap***

ABSTRACT

The syllabus, which outlines learning objectives, evaluation standards, and course policies, acts as the main agreement between instructors and students in university courses. However, syllabi are now treated as static materials, usually shared once via learning management systems as Word files and hardly reviewed. Poor information design in course materials increases student anxiety and prompts recurrent questions to instructors, according to research on the use of educational technology (Nilson & Goodson, 2021). Studies on syllabus accessibility show that students find it difficult to locate important material during stressful times, while instructors struggle with version control and lack visibility into how students are interacting with course policies (Harrington & Gabert-Quillen, 2015). This project uses an AI-supported online platform to address these documented issues by converting the syllabus from a static document into an intelligent, interactive resource.

The process is divided into two stages and uses a user-centered design approach. Phase 1 (Fall Term) starts with a thorough examination of the literature that looks at information retrieval interfaces in academic settings, AI-powered teaching tools, and current syllabus management systems. After that, 15–20 undergraduate students and 4-6 instructors from various disciplines will participate in semi-structured interviews to determine feature needs, usage trends, and pain points. Thematic analysis will be used to extract system requirements and user demands from the data. These results will guide the development of low-fidelity wireframes and high-fidelity interactive prototypes using Figma, which will undergo iterative usability testing with 8-10 participants. Phase 2 (Spring Term) concentrates on implementation using React for frontend development and Python-based NLP models for AI features, such as semantic search, deadline extraction, and intelligent summarization. The system's accuracy and task completion time will be compared to those of conventional PDF syllabi through controlled user testing.

An agile framework with two-week sprints is used in project management. Fall term milestones include: completing the literature review (Week 4), gathering and analyzing interview data (Weeks 5–8), defining requirements (Week 10), and developing prototypes (Weeks 11–14). Spring term deliverables consist of MVP development (Weeks 1–8), user testing (Weeks 9–11), and final documentation (Weeks 12–14). Risk mitigation techniques include maintaining a feature prioritization matrix to control scope and scheduling weekly advisor check-ins to ensure compliance with academic standards.

Anticipated outcomes include: (1) a functional web-based prototype showcasing essential features such as AI-powered search, personalized deadline tracking, and version-controlled syllabus updates; (2) comprehensive user testing results with quantitative metrics on information retrieval efficiency; (3) technical documentation describing system architecture and AI model integration; and (4) a research paper suitable for submission to educational technology conferences, providing empirical evidence on AI applications in course management systems.

Key Words
syllabus,usability, artificial intelligence, academic communication,digital learning

TABLE OF CONTENTS

LIST OF TABLES

LIST OF FIGURES

LIST OF ABBREVIATIONS

IoT	Internet of Things
M2M	Machine-to-Machine
IEEE	The Institute of Electrical and Electronics Engineers

1. Background and Motivation

1.1 Background:

In higher education, the syllabus is vital since it creates expectations and defines the general format of a course. Yet, in spite of its importance, it can be seen as a static document that is given out at the start of the semester and is rarely ever reviewed again.

Students sometimes have to navigate long, text-heavy documents in order to access important data like deadlines, assessment criteria, or grading standards. This procedure can be unproductive and mentally draining especially for students who are enrolled in several classes at once. Teachers also have to deal with the fact that even little changes to the syllabus typically require rearranging materials and making more announcements, with little indication of whether the new information has been observed or understood.

The syllabus itself is still largely passive, despite the growing popularity of online educational and communication tools in higher education. There is still a gap between current digital capabilities and normal course-level practices due to the lack of systems that support the syllabus as a constantly changing, continuously accessible, and user-friendly part of the educational procedure.

1.2 Scope and Problem Statement:

This project examines how syllabi are used by students and instructors in higher education and explores ways to improve this experience through a dedicated digital system. The syllabus is considered as an academic communication tool rather than a technical or administrative document. In current practice, syllabi are mostly shared as static files on general communication platforms. Over time, these files become difficult to track and update, which often leads to confusion and repeated clarification during the semester.

The project does not aim to redesign or replace existing learning management systems. Instead, it focuses on identifying usability-related issues in current syllabus practices and evaluating potential improvements from a user experience perspective.

By concentrating on accessibility, clarity, and ease of use, the study seeks to support more effective communication between instructors and students.

1.3 Project Objectives:

The main objective of this project is to examine and improve the syllabus experience in higher education by focusing on usability and user experience aspects.

To identify common difficulties faced by students and instructors in accessing and using syllabus information.
To analyze current syllabus practices from a user experience perspective.
To define user needs related to syllabus clarity, accessibility, and updates.

To explore how AI-supported features can assist users in understanding and navigating syllabus content.
To evaluate potential improvements for syllabus management based on usability and UX principles

1.4 Motivation and Significance:

This projectis important because the syllabus is one of the main documents students and instructors use duringacourse. When this document is difficult to follow, it often causes confusion and repeatedquestions about the courses. Student benefit from having clearer syllabus information, especiallywhen they are taking several courses at the same time. Instructors also benefit, since clearerinformation reduces the need for constant explanations, follow-up messages.

For thisreason, improving how syllabus information is presented and used can make everyday academiccommunication easier for both students and instructors.

1.5 Limitations

This project is limited to the analysis of current syllabus usage, the identification of user needs, and the evaluation of syllabus-related usability aspects within higher education. The primary focus is on students and instructors as the main user groups. The study explicitly excludes the development of a full learning management system, as well as long-term deployment, maintenance, or large-scale implementation.

1.5.1 Technical Constraints

The project is conducted within a limited academic timeframe. As a result, the technical scope remains restricted and does not include complex system development or advanced engineering solutions.

The tools used during the project are limited to widely available and commonly used software platforms. Advanced or specialized tools are not considered. AI-related components are intentionally kept simple and are used only as supportive elements, such as assisting with information organization or navigation. Model training, large-scale data processing, and advanced artificial intelligence techniques are outside the scope of this study.

1.5.2 Realistic Constraints

The project is shaped by several realistic constraints related to economic, ethical, and social factors.From an economic perspective, the study is conducted within the boundaries of an academic projectand does not have a dedicated budget. Therefore, commercial software, advanced AI infrastructure, and paid data sources are not used. The project relies on freely available tools and limited technical resources.

Social factors also play a role in shaping the project. Students and instructors have different expectations, habits, and levels of engagement when interacting with syllabus information, which affects how usability issues are identified and evaluated. In addition, the study focuses specifically on academic communication practices and does not address broader institutional, administrative, or policy-level concerns.

2. Literature Review / State of the Art

UX & User-Centered Design in Educational Systems

Previous research highlights the significance of user-centered and usability-oriented strategies in designing university and e-learning platforms. Alao et al. assert that involving users in the design phase enhances the clarity and usability of systems [1]. In a similar vein, Alqurni notes that ineffective interface design and complicated navigation diminish user satisfaction within e-learning environments [3]. Recently, Maskur and Syarief performed a thorough review of usability and user experience evaluation techniques for websites of higher education institutions, revealing advancing trends in UX assessment methodologies [10]. While these studies underscore the importance of usability in academic systems, they primarily concentrate on general platforms rather than syllabus-specific use cases.

E-learning Platforms and Digital Learning Environments

The effects of e-learning platforms on sustainability and accessibility in higher education have been widely studied. Gavrus et al. analyze the role of such platforms across different regions and demonstrate that digital systems support flexible learning environments [2]. Through bibliometric research, Vergara et al. further examine how AI affects learning management systems, demonstrating the increasing incorporation of AI technologies in educational platforms [6]. However, these studies mostly focus on platform-level advantages rather than how users interact with specific academic documents, such as syllabi, in their day-to-day academic practice.

AI-Supported Learning and Curriculum Development

The application of AI-based tools in higher education has been examined from a variety of perspectives in recent literature. The role of AI-supported learning systems in personalization and learningassistance is reviewed by Luo et al. [4]. In their analysis of learners' expectations for AI-poweredlearning, Kim et al. stress the importance of meaningful human–AI interactions and clarity [5]. Fromthe perspective of curriculum development, Walter offers a case study on the use of language-generating AI in educational planning [7], while Kasztelnik explores advances in creating educational content with AI support for 21st-century learners [9]. The broader context of software-assisted educational management is illustrated in Lam et al.'s survey of software tools for process integration and optimization [8]. Despite these contributions, the use of AI as a supportive tool for organizing and presenting syllabus content from a user experience perspective is rarely addressed in the literature.

Research Gap

Although the growing research on university information systems, e-learning platforms, and AI-driven instructional technologies is increasing, a significant portion remains focused on a broad system level. Most existing studies prioritize aspects such as usability, accessibility, or learning outcomes instead of analyzing the syllabus as an independent academic document utilized during the course. In particular, there has been insufficient research on the relationships between students and instructors regarding syllabi, nor on the potential for AI to enhance the accessibility and clarity of syllabus information. The present project is based on this deficiency in the available literature

3. Methodology and Technical Approach

3.1 Overall Approach and Tasks:

In order to create and implement AI-supported syllabus software that improves syllabus usability for both teachers and students, this project uses an iterative, user-centered development technique. The method ensures that user needs directly impact system functionality by integrating software development, UX design, and user research. The following are the project's primary tasks:

1.Definition of the Problem and Review of Literature:
Evaluating scholarly research on AI-supported educational systems and UX design, as well as identifying issues with the use of traditional curricula.
2.Needs Analysis and User Research:
Gathering qualitative information from instructors and students via surveys and interviews in order to comprehend usability concerns and expectations.
3.Definition of Requirements:
Converting user requirements into both functional and non-functional system specifications.
4.Interface and System Design:
Utilizing usability concepts when designing user interfaces and system architecture.
5.AI Integration and Implementation:
Python is used in the software's development, and AI language model APIs are integrated to facilitate interaction and comprehension of the curriculum.
6.Testing and UX Assessment:
Testing usability and evaluating UX data that has been gathered.
7.Reporting and Documentation:
Recording conclusions, outcomes, and recommendations for future development.

3.1.2 User Research and Needs Analysis

In order to design a more usable and effective AI-supported syllabus system, a user-centered research study was conducted to understand the real problems experienced by both students and instructors.

Participants

The study involved:

15–20 undergraduate students from different departments such as Industrial Engineering, Software Engineering, and Computer Engineering.

4–6 instructors who are actively teaching undergraduate courses.

The participants were selected to represent different user perspectives and academic roles.

Data Collection Process

Data was mainly collected through ***short, structured forms and brief interviews***.

Most of the sessions were conducted ***face-to-face***, while a smaller portion was completed ***online*** for convenience.

The form required approximately ***5 minutes*** to complete and focused on practical usage issues rather than technical system details.

Sample Questions

For students:

Which information do you check most frequently in the syllabus?

Which parts of the syllabus are usually unclear?

Which features would improve your syllabus experience?

For instructors:

How do you currently prepare and share your syllabus?

How often do you update your syllabus during the semester?

Which parts of the syllabus do students struggle with the most?

Which features would you expect from a web-based syllabus system?

Which AI features would you find helpful?

Main Findings – Students

Students reported that they mostly struggle with:

Tracking ***exam dates and assignment deadlines***

Understanding ***weekly plans and grading systems***

The most frequently requested features were:

Search within the syllabus

Mobile-friendly interface

Automatic reminders and notifications

Many students also stated that having an AI-based assistant to answer frequently asked questions would reduce confusion and save time.

Main Findings – Instructors

Instructors indicated that students mainly have difficulties with:

Weekly schedule and course flow

Grading policies

Exam and assignment dates

Course rules and learning outcomes

A common problem for instructors is the need to ***re-upload the entire document*** after every small update.

They also mentioned that students repeatedly ask similar questions, mostly related to quizzes and exams.

Expected features from a web-based system include:

Easy syllabus creation and editing

Weekly content management

File upload support (PDF, slides)

Announcements and student feedback collection

Advanced search functionality

Regarding AI-based support, instructors found the following features useful:

Automatic syllabus formatting

Detection of missing syllabus components

Automatic answering of student questions

Some instructors also suggested that the system could make course selection and syllabus access easier (e.g., students not being able to see a syllabus unless they are officially enrolled) and mentioned the potential value of a digital twin for course management.

Identified User Needs

Based on the collected data, the key user needs were identified as:

Faster and clearer access to critical course information

Reduced cognitive effort when interacting with syllabus content

Easier syllabus updating for instructors

Consistent communication of changes to students

Intelligent AI-supported assistance for both students and instructors

These needs directly informed the functional and non-functional requirements of the proposed system.

3.2 System Design / Architecture:

To guarantee simplicity, maintainability, and ease of development, the system is built as a desktop program with a modular architecture. There are three primary parts to the architecture:

•User Interface Layer:

Using organized screens and forms, this Tkinter-implemented layer enables instructors and students to engage with the syllabus content.

•Application Logic Layer:

This Python-developed layer controls user interactions, syllabus operations, and AI service communication.

•Data Layer:

User data, interaction records, and syllabus data are stored in a relational database (MySQL).

Features like syllabus summarizing, explanation, and content support are made possible by the integration of AI language model services through API calls within the application logic layer.

System Requirements

This section outlines the core requirements of the AI-supported syllabus management platform. The requirements are defined to guide system design and implementation while maintaining usability, reliability, and ethical use of AI technologies.

Functional Requirements

•The platform is expected to support instructors in uploading and managing syllabus content for individual courses.

•Syllabus information should be presented in a structured and easy-to-navigate format to improve clarity and accessibility.

•Version control is required to ensure that users always access the most up-to-date syllabus content.

•Users should be informed when changes are made to syllabus content to prevent confusion or outdated information.

•An AI-supported chat interface is required to allow users to ask questions related to syllabus content.

•AI-generated responses should rely strictly on the approved syllabus material to ensure accuracy and consistency.
•The platform should allow efficient searching of syllabus content through keywords or natural language queries.

Non-Functional Requirements

•The system should be intuitive and require minimal effort for new users to become familiar with its features.
•AI-supported interactions should provide consistent and reliable responses across different use cases.
•Response times for search and chat functions should be reasonable to support smooth user interaction.
•The platform should be accessible on commonly used devices and adaptable to different screen sizes.
•Basic data privacy principles and ethical AI usage guidelines should be followed throughout system operation.

3.3 Tools, Technologies, and Standards:

The following technologies and tools are used in the project:
•Programming Language: Python
•Frontend: Tkinter (desktop graphical user interface)
•Backend & Database: MySQL relational database; Python-based backend logic
•AI & Data Processing:
•Python data handling libraries
•AI language model APIs (like the OpenAI, Llama API)
•Development Environment:
•Visual Studio Code
•Design and Documentation:
•UML tools (draw.io, Lucidchart)
•Microsoft Word for project documentation
•Standards and Practices:
•Modular software design
•Basic data privacy and ethical AI usage principles

3.4 Data Collection and Analysis (if applicable): The qualitative and quantitative data collected from user research and usability testing were analyzed using descriptive analysis methods. The responses from students and instructors were grouped based on recurring themes such as accessibility, clarity, update frequency, and communication efficiency. As a result of this analysis, the following user needs were identified.

The main areas of data collecting for this project are usability assessment and user research.

•User Research Data:

To determine usability issues and expectations, instructors and students were surveyed and interviewed.

•System Interaction Data:

Basic usage information gathered during testing, such as the frequency of feature usage and user comments.

•UX Evaluation Data:

Task completion rates and subjective user satisfaction from usability tests.

Descriptive approaches are used to analyze collected data in order to assess system usability and pinpoint areas that require improvement.

Based on the analysis of the collected data, several key user needs were identified. These needs are summarized below.

Student Needs

•Students need a clear and easily accessible way to reach syllabus information throughout the semester.

•Students need to quickly find critical details such as deadlines, grading criteria, assessment dates and times, and attendance rules without navigating long documents.

•Students need access to course syllabi before course registration in order to make informed course selection decisions.

•Students need timely and visible notifications when changes are made to the syllabus, including updates related to assessment schedules.

•Students need a system that reduces uncertainty and confusion by presenting syllabus content in a structured and readable format.

•Students need an AI-supported chat interface that allows them to ask syllabus-related questions in natural language and receive immediate, accurate answers related to course requirements and schedules.

Instructor Needs

•Instructors need a practical way to add, edit, and update assessment schedules within the syllabus without creating multiple document versions.

•Instructors need assurance that students are accessing the most recent and correct version of the syllabus.

•Instructors need to reduce repetitive questions related to course policies, grading, deadlines, and schedules.

•Instructors need clearer communication channels that help align student expectations with course rules.

Communication and System-Level Needs

•Both students and instructors need a shared platform that supports consistent and transparent academic communication.

•The system needs to strengthen the connection between students and instructors by minimizing misunderstandings related to syllabus content.

•The platform needs to support continuous access to syllabus information rather than one-time document distribution.

•AI-supported features need to operate strictly based on official syllabus content to ensure reliability and trust

3.5 Implementation Plan:

The framework of the implementation process consists of the following steps:

1. Completing functional needs based on the results of user research.
2. Design of Interfaces Make interface prototypes and wireframes using UX design tools.
3. Development of Systems: The desktop application is implemented using Tkinter, MySQL, and Python.
4. AI Integration: Using AI language model APIs to make syllabus assistance easier.
5. Testing and UX Assessment: collecting user feedback and performing usability evaluations.
6. Full Reporting and Record-Keeping compiling the final project documentation and evaluation results.

Wireframe Designs

Login/ Role Selection Screen

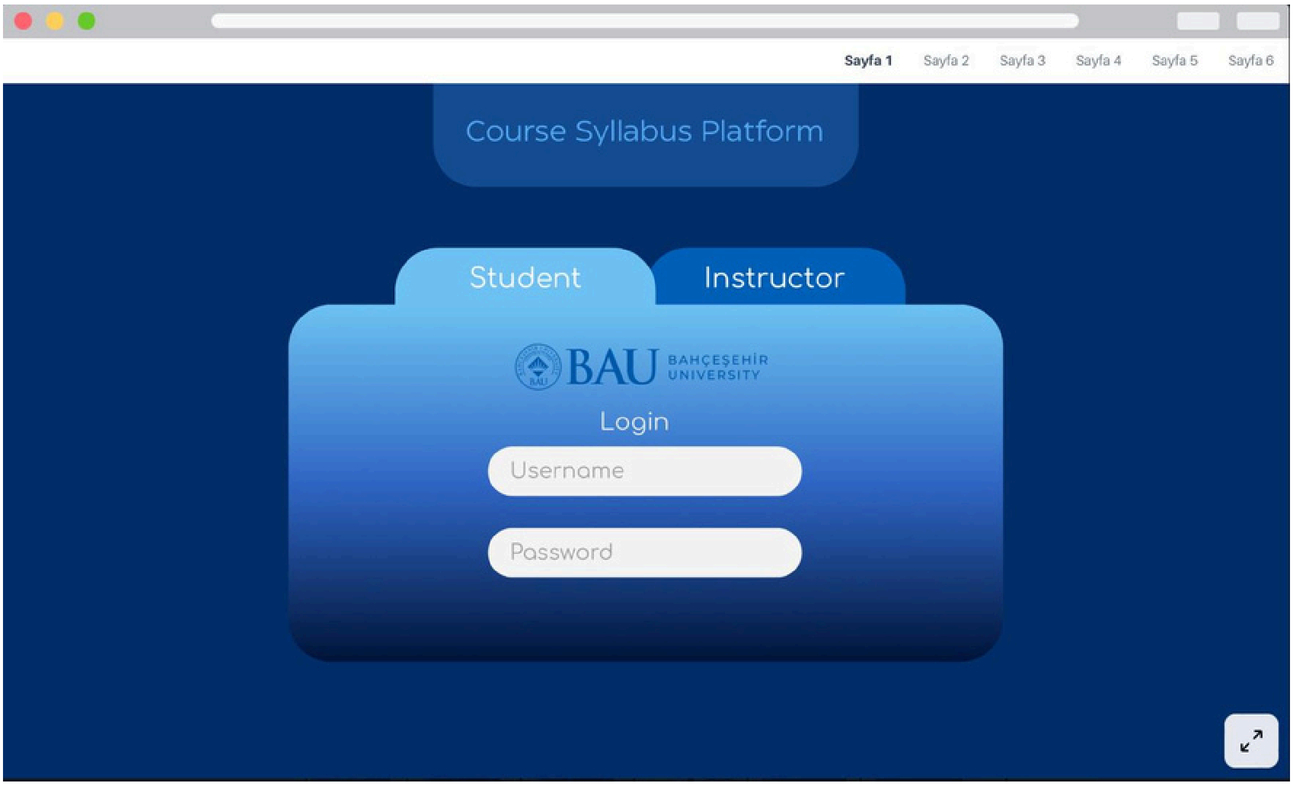


Figure 1. Login / Role Selection Screen Wireframe

Student Dashboard

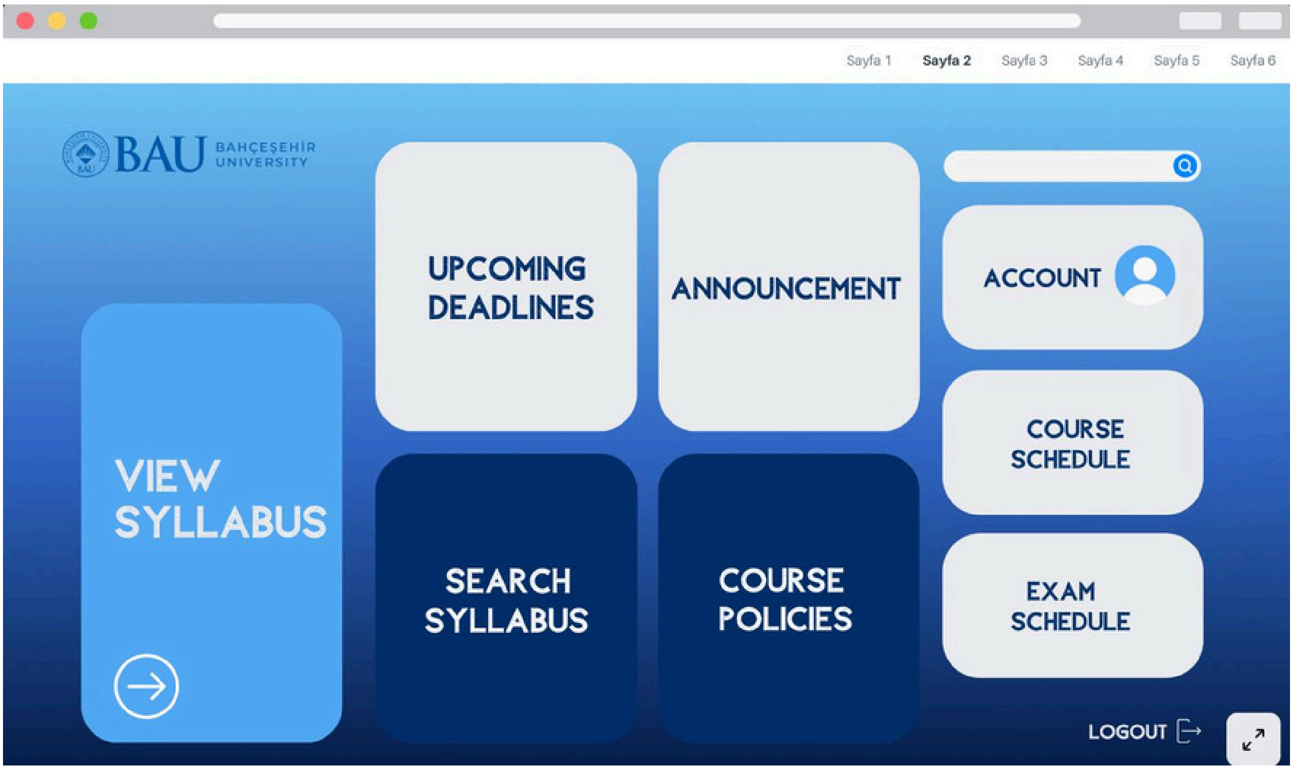


Figure 2. Student Dashboard Interface of the AI-Supported Syllabus Platform

Instructor Dashboard

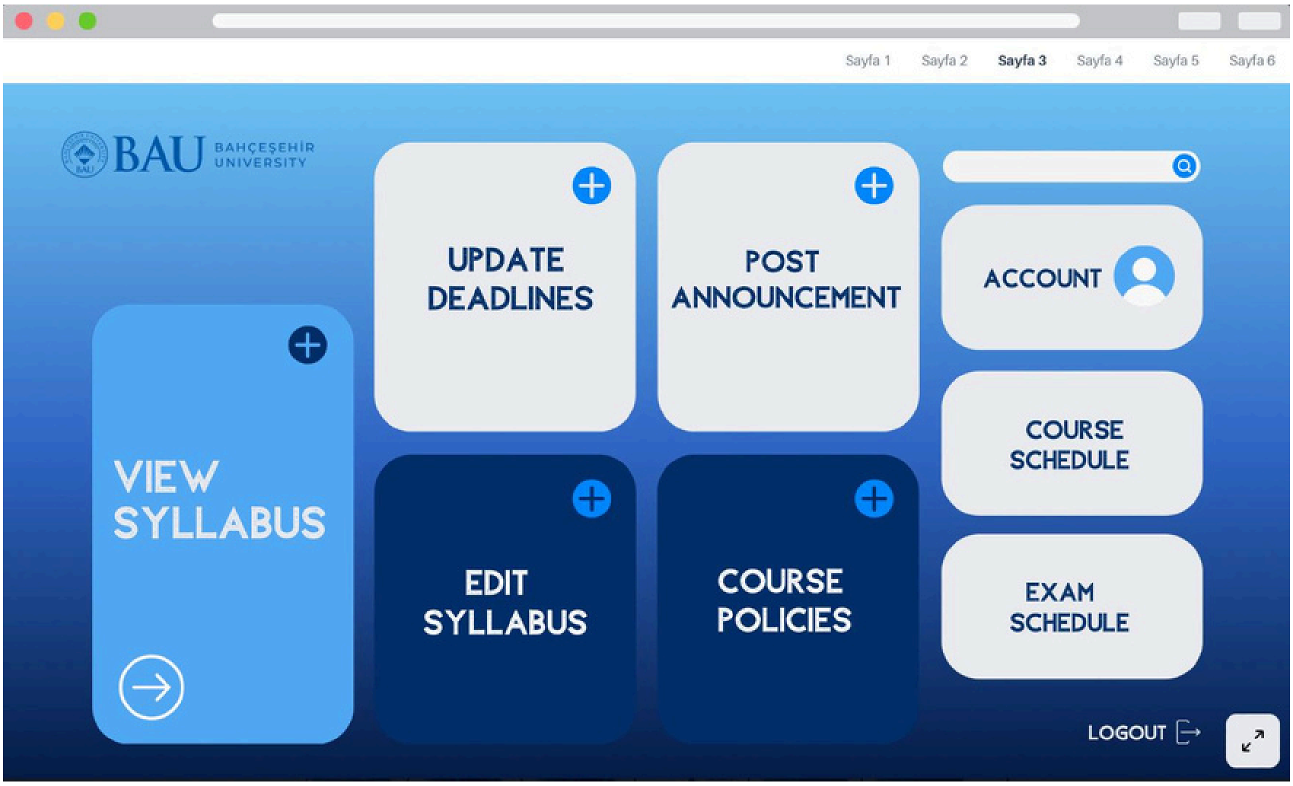


Figure 3. Instructor Dashboard Interface of the AI-Supported Syllabus Platform

Syllabus View Page

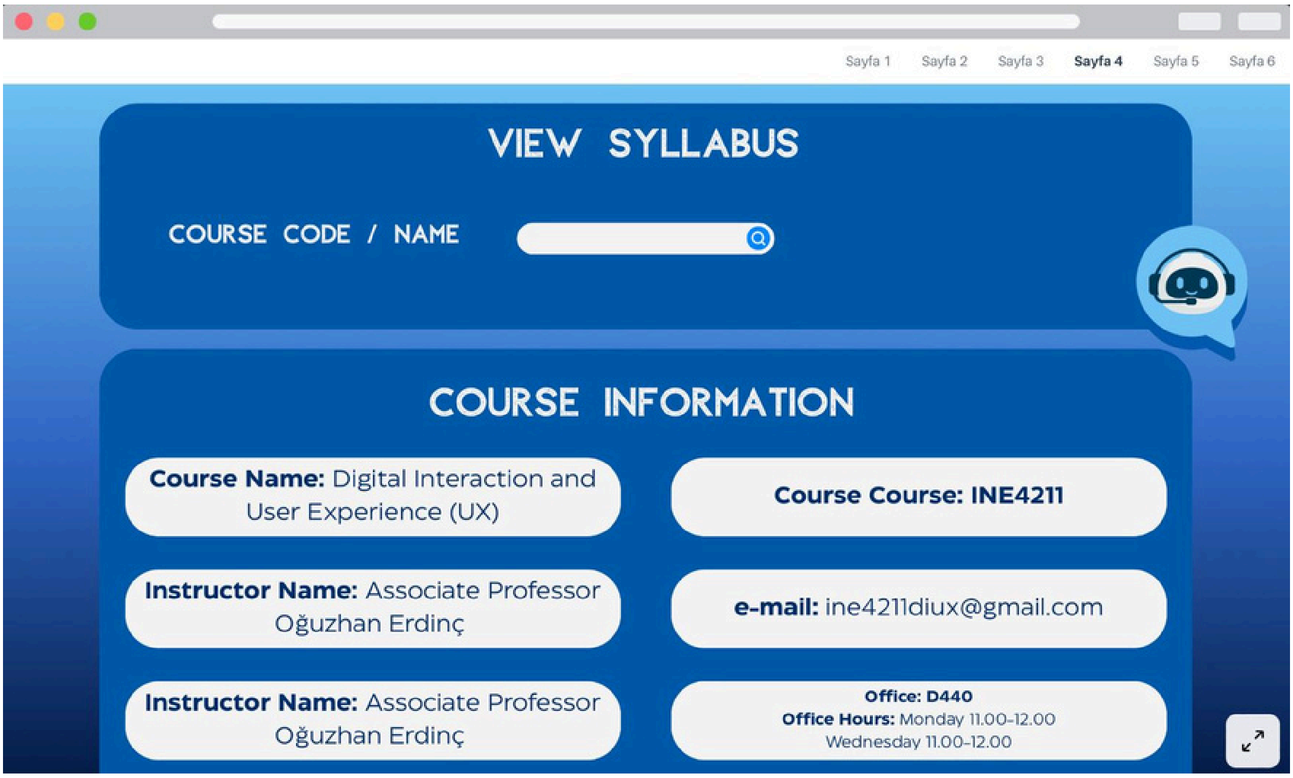


Figure 4. Syllabus View Page ,

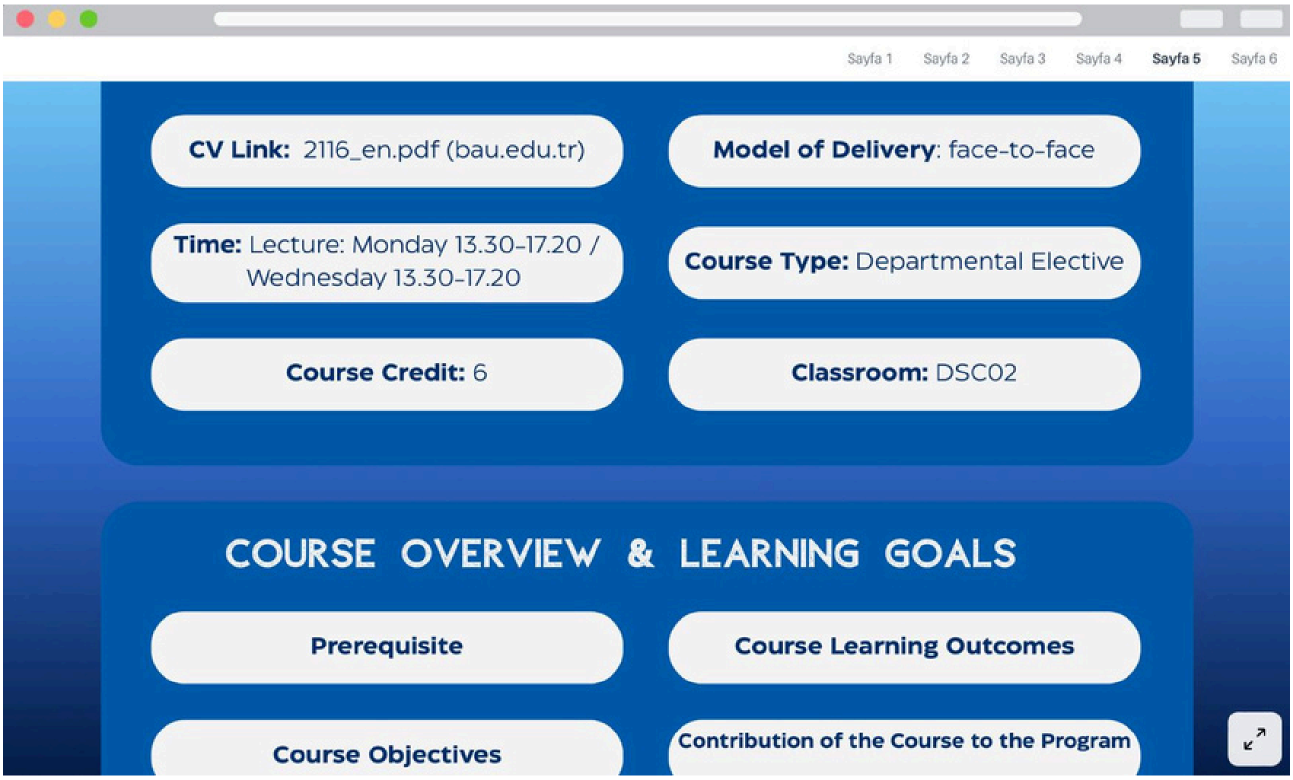


Figure 5. Syllabus View Page ,

AI- Supported Syllabus Search

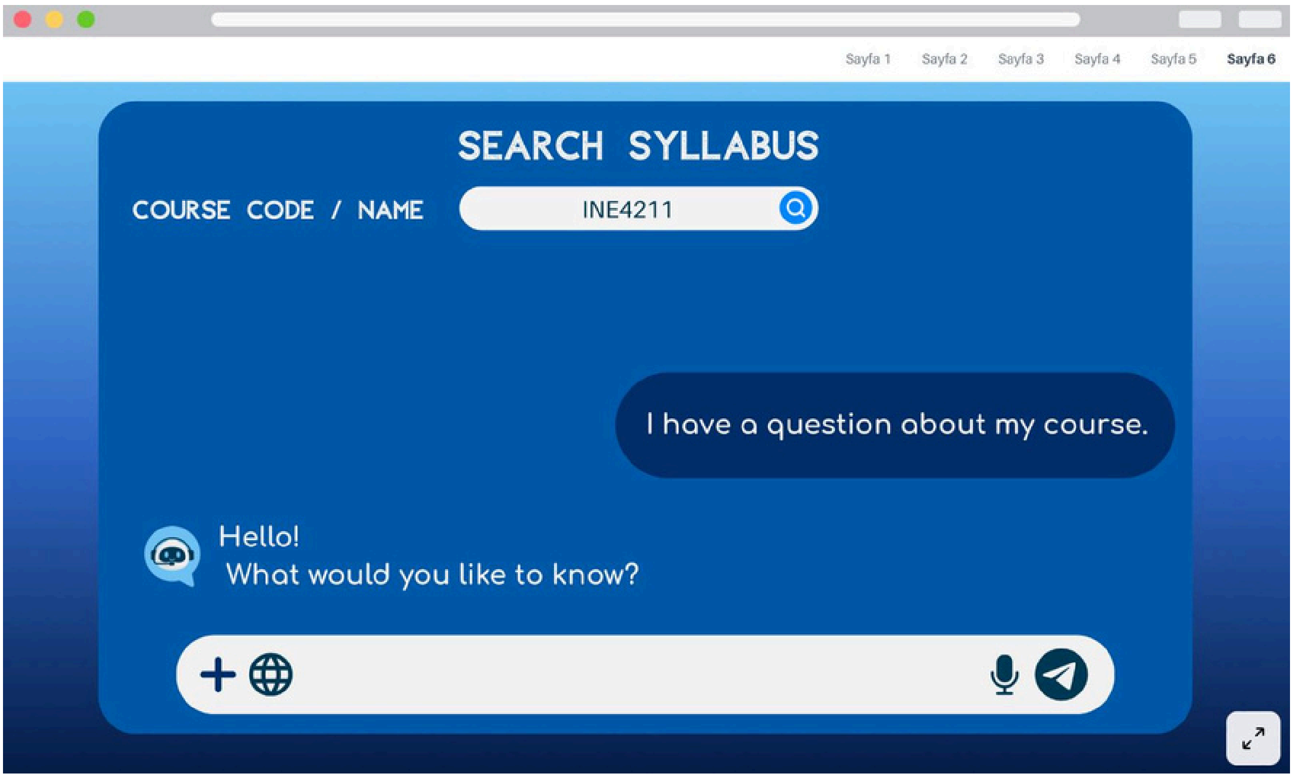


Figure 6. AI-Supported Syllabus Search

DesignConsiderations

The interface design was shaped by basic user experience (UX) principles together with selected ***Gestalt design laws***. While developing the wireframes, the main intention was not to create a visually complex interface, but to support users in understanding and navigating academic information more easily, especially during busy periods such as exams.As shown in Figures 1–6, these principles were reflected in the layout and grouping of interface components in the wireframe designs.

One of the principles reflected in the design is the ***Gestalt law of proximity***. Information that belongs together, such as course content, grading criteria, and syllabus-related details, was placed close to each other. This allows users to perceive related elements as a whole without having to consciously search for connections between different parts of the screen.

The law of similarity was also considered when designing interface components. Elements that serve similar functions, including buttons and content blocks, were designed using similar visual styles. This consistency helps users become familiar with the system more quickly and reduces confusion, particularly for first time users.

Another design decision was influenced by the ***Gestalt law of continuity***. Content was aligned in a clear and structured flow, making it easier for users to follow information in a natural reading order. This approach supports quick scanning rather than forcing users to read long sections of text in detail.

Beyond Gestalt principles, general usability considerations were taken into account. ***A clear visual hierarchy*** was used to highlight important information such as deadlines and announcements. At the same time, unnecessary visual elements were deliberately avoided in order to reduce cognitive load and keep the interface simple and functional.

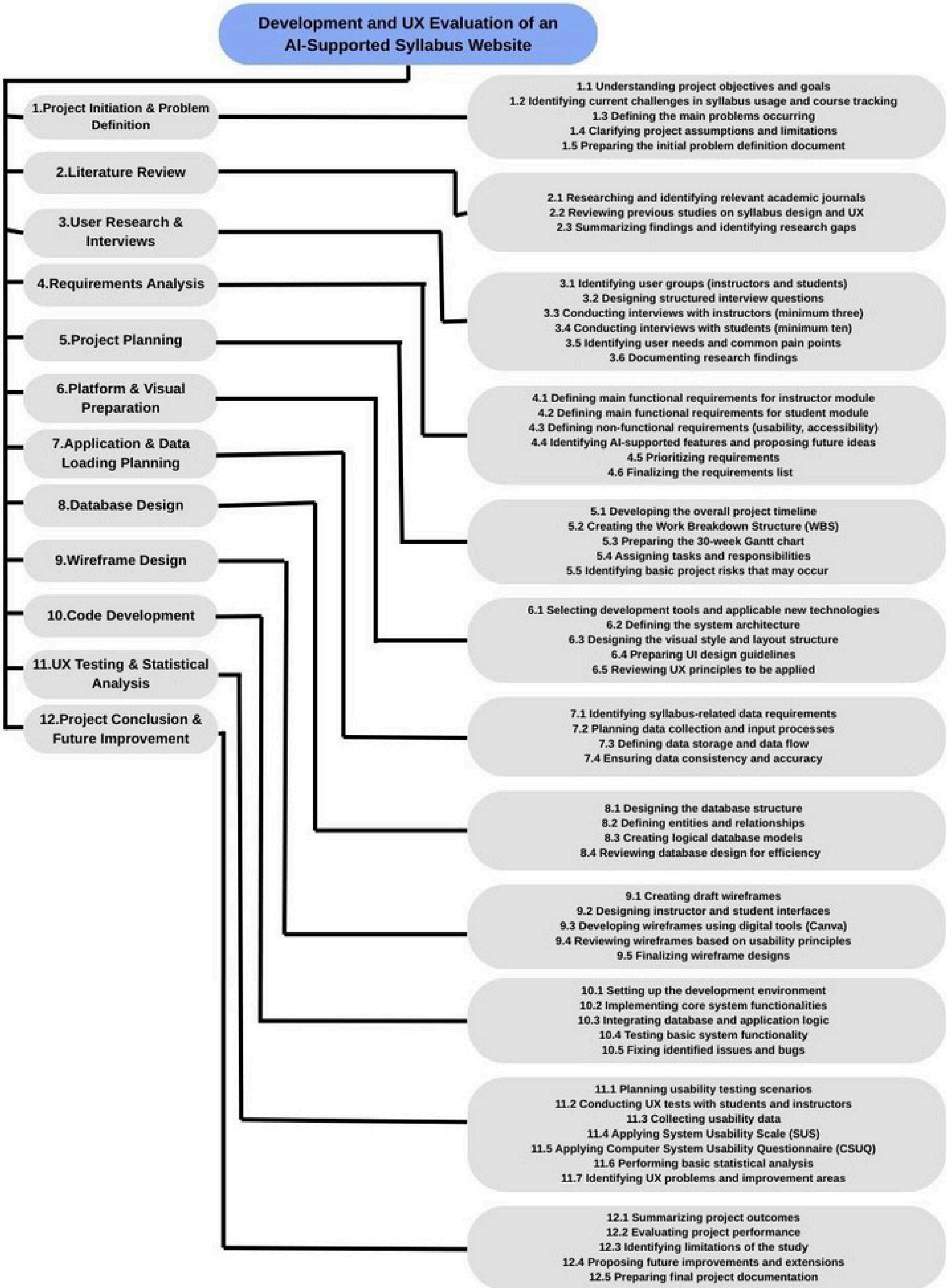
Finally, the wireframes were intentionally kept at a low fidelity level. This allowed the design to remain flexible and open to improvement. As the system evolves and user testing is conducted, the interface can be revised based on real feedback, ensuring that the final design reflects actual user needs rather than initial assumptions.

Tools Used The wireframes were created using ***Canva***, which was preferred for its simplicity and ease of use duringthe early design stage. Canva allowed quick visualization of interface ideas and supported flexiblerevisions as the design evolved.

4. Work Plan and Schedule

4.1 Work Breakdown Structure

Project:Development and UX Evaluation of an AI-Supported Syllabus Website
Duration: 30 Weeks



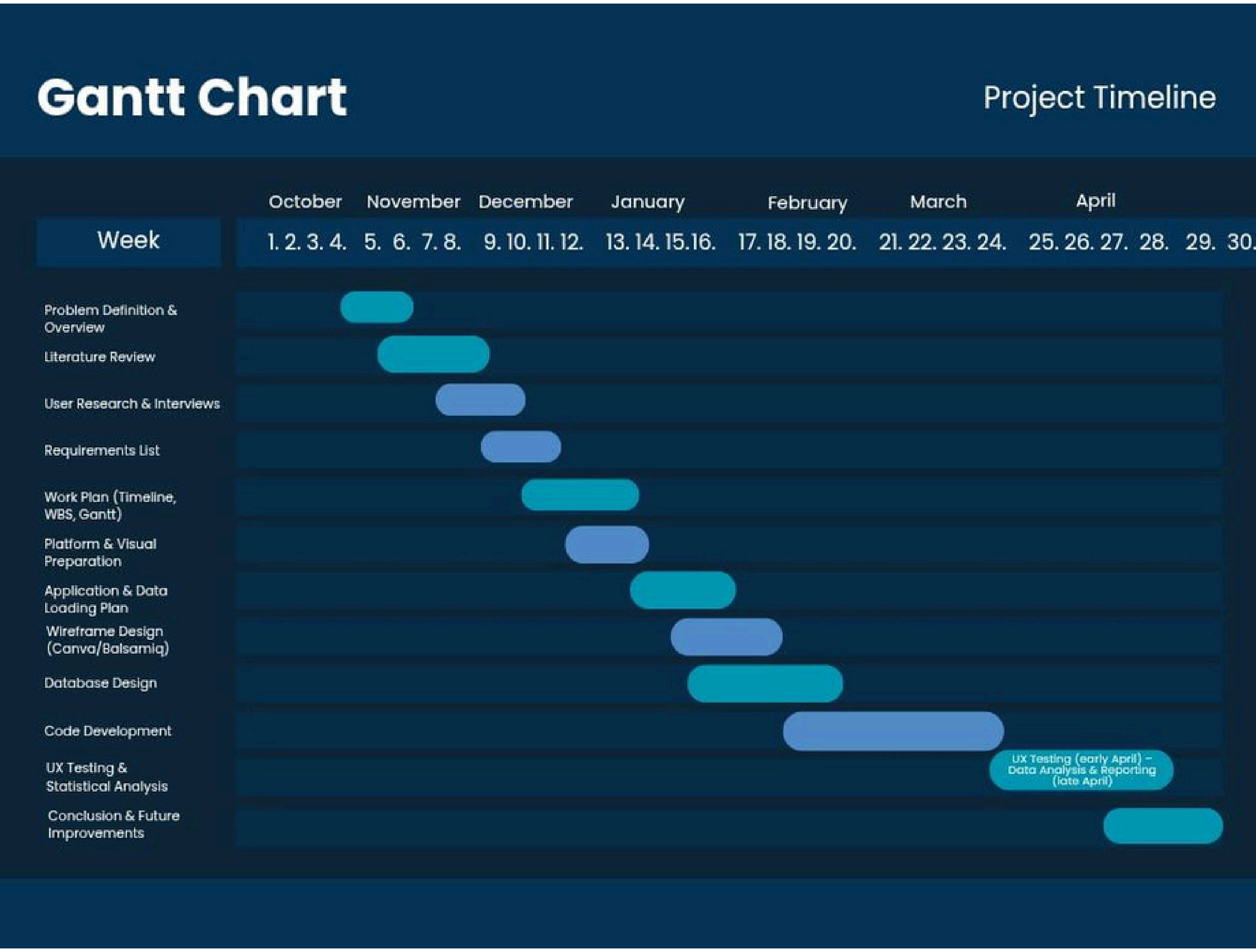
4.2 Gantt chart

The Gantt chart is used to show the project timeline in a clear way and to reflect how the tasks are distributed over the semester. The project is planned as a 30-week process and includes several phases that overlap at certain points, which is common in software development and UX-oriented projects.

At the beginning of the timeline, the focus is on project initiation, problem definition, and the literature review. Starting with these tasks helps the team develop a shared understanding of the project scope and academic background before moving into design-related work. Once this foundation is established, user research and requirement-related activities take place. Feedback gathered from students and instructors at this stage is used to shape both system requirements and design choices.

After the requirements become clearer, planning activities such as finalizing the WBS and preparing the Gantt chart are completed. This order is preferred so that the schedule is based on realistic task definitions rather than assumptions. During the middle part of the project, visual design, application planning, and database design are carried out. These activities help bridge the gap between early conceptual work and the implementation phase. A relatively large portion of the timeline is allocated to code development. This is mainly because the system is developed in an iterative manner and includes multiple components, such as separate interfaces for students and instructors, as well as AI-supported features. Following development, the project moves into usability testing and basic statistical analysis. At this stage, usability data is collected and evaluated using tools such as SUS and CSUQ. Allowing enough time for testing makes it easier to observe usability problems and interpret the results in a structured way.

In the final weeks, the focus shifts to evaluation, documentation, and discussing possible future improvements. This phase allows the team to review the overall outcomes, acknowledge limitations, and consider how the system could be extended in later studies. Overall, the Gantt chart helps connect the project tasks with the planned timeline and the deliverables defined in the WBS, without treating the process as a strictly linear sequence.


Gantt Chart
Project Timeline
October
November
December
January
February
March
April
Week
1. 2. 3. 4. 5. 6. 7. 8. 9. 10. 11. 12. 13. 14. 15. 16. 17. 18. 19. 20. 21. 22. 23. 24. 25. 26. 27. 28. 29. 30.
Problem Definition & Overview
Literature Review
User Research & Interviews
Requirements List
Work Plan (Timeline, WBS, Gantt)
Platform & Visual Preparation
Application & Data Loading Plan
Wireframe Design (Canva/Balsamiq)
Database Design
Code Development
UX Testing & Statistical Analysis
UX Testing (early April) – Data Analysis & Reporting (late April)
Conclusion & Future Improvements

5. Required Resources

5.1 Hardware

No specialized hardware is required for this project. Development, testing, and deployment can be carried out using standard personal computers or laptops with internet access.

5.2 Software

The following software tools and technologies are required:

Development Tools

- o Visual Studio Code

Backend & Database

- o Relational Database Management System (MySQL)
- o Backend framework (Python)

Frontend

- o Tkinter or HTML,CSS will decide later based on the outcome

AI & Data Processing

- o AI language model APIs (OpenAI API ,Llama or similar)
- o Python

Documentation & Design

- o UML modeling tools
- o Project documentation tools (Word)

5.3 Facilities and Services

Cloud services for hosting and testing (e.g., AWS, Azure, or similar)

External AI services via API access

5.4 Estimated Budget

The project is expected to incur ***minimal cost***. Most tools used are open-source or provided through educational licenses.

Cloud hosting (optional): Low-cost or free student tier

AI API usage: Limited usage within free or trial quotas

Total estimated budget: ***Low / negligible***

6. Risk Assessment and Mitigation

In this project, several technical, scheduling, and user-related risks may affect the development and evaluation of the AI-supported active syllabus system. These risks were identified at an early stage in order to reduce their potential negative impact on the project outcomes. A risk matrix approach was used to assess each risk based on its ***probability*** and ***severity***, and appropriate mitigation actions were defined accordingly.

Table 5. Risk matrix

RISK LEVEL		**Severity** of the event on the project success		
		Minor	Moderate	Major
Probability of the event occuring	Unlikely	VERY LOW	LOW	MEDIUM
	Possible	LOW	MEDIUM	HIGH
	Likely	MEDIUM	HIGH	VERY HIGH

VERY LOW	This event is very low risk and so does not require any plan for mitigation. In the unlikely event that it does occur there will be only a minor effect on the project.
LOW	This event is low-risk; a preliminary study on a plan of action to recover from the event can be performed and noted.
MEDIUM	This event presents a signficant risk; a plan of action to recover from it should be made and resources sourced in advance.
HIGH	This event presents a very signficant risk. Consider changing the product design/project plan to reduce the risk; else a plan of action for recovery should be made and resources sourced in advance.
VERY HIGH	**This is an unacceptable risk. The product design/project plan must be changed to reduce the risk to an acceptable level.**

Table 6. Riskassessment

Failure event	Probability	Severity	Risk level	Plan of action
Difficulty finding enough students and instructors for interviews and usability tests	Possible	Moderate	MEDIUM	Contact participants early, use online interviews if needed, and keep backup participants available.
Instructors not willing to share detailed syllabus information	Possible	Major	HIGH	Prepare short and clear data collection forms, anonymize responses, and explain the academic purpose of the study.
Delays in prototype or wireframe development	Possible	Major	HIGH	Limit the prototype scope to core functions, prioritize essential features, and divide tasks clearly among team members.
AI-supported features taking longer than expected to implement	Likely	Moderate	HIGH	Start with simple rule-based or semi-automated features and improve AI components incrementally if time allows.
Low usability scores (SUS / CSUQ) in initial tests	Possible	Moderate	MEDIUM	Analyze UX results carefully, identify major usability problems, redesign critical screens, and conduct follow-up tests if possible.
Schedule conflicts among team members	Possible	Moderate	MEDIUM	Use a shared project schedule, assign clear responsibilities, and hold regular coordination meetings.
Data loss or version conflicts during development	Unlikely	Major	MEDIUM	Use cloud storage and version control tools, and perform regular backups throughout the project.

7. *Expected Outcomes and Evaluation*

7.1 Expected Outcomes:

The design and development of a conceptual framework and/or functional prototype that redefines the course syllabus as an active, user-oriented part of the academic process is the project's expected result. Instead of being limited to a static document format, the recommended strategy will show how syllabus content can be organized, accessible, and updated dynamically during a semester.

The following will be key outcomes:

-A system model or prototype that demonstrates the active syllabus concept,

-A methodical approach to arranging syllabus data (such as due dates, procedures, and rules for assessments),

-A written report that details the design justification, problem analysis, and suggested solution, usage scenarios or student-teacher interaction processes, if appropriate.

7.2 Success Criteria:

In this project, the success of the proposed active syllabus system is evaluated primarily from a user experience perspective. Rather than focusing only on technical performance or ideal usability scores, the main objective is to understand whether the system meaningfully supports students and instructors in managing course-related information.

Since this is the first version of the system, success is not defined by achieving perfect results, but by identifying how well the system responds to user needs and where improvements are required. Both quantitative indicators and qualitative user feedback are considered together to form a realistic evaluation.

Quantitative Success Indicators

From a quantitative perspective, the system will be considered successful if the evaluation reveals positive usability-related trends, such as:

- Users being able to locate syllabus information more efficiently compared to traditional static syllabus documents.
- Reasonable task completion rates during basic syllabus-related activities, such as checking deadlines or course rules.
- SUS and CSUQ results that provide meaningful insights into usability, even if scores indicate areas for improvement.
- Observable engagement with the system throughout the semester, such as repeated access to course topics or important dates.

It is acknowledged that usability scores may not be high in the first evaluation, and such results are considered valuable for guiding further development.

Qualitative Success Indicators

In addition to numerical outcomes, qualitative feedback plays an important role in defining success. The system will be viewed positively if:

- Students and instructors describe the interface as understandable, structured, and easier to follow than existing syllabus formats.
- Users report reduced confusion when accessing course information, even if some issues remain.
- Feedback highlights specific usability problems, suggestions, or unmet needs that can inform future design improvements.
- Critical usability issues are identified and documented during user testing sessions.

Overall, if the evaluation helps reveal both strengths and weaknesses of the system in a realistic way, the project will be considered successful.

7.3 Evaluation Methods

The system will be evaluated using a combination of user testing, SUS, and CSUQ questionnaires. These methods are used to collect both qualitative and quantitative feedback regarding usability, clarity, and user satisfaction.

The evaluation plan is now described as a beginning, flexible framework rather than a set structure. In order to guarantee that evaluation is viewed as a crucial part of the design and development process rather than as a last stage, this early definition was created. The evaluation scenarios, task definitions, and questionnaire items will be continually improved as the system develops and new features are added to better reflect actual interactions between users and system functionality. This adaptable strategy guarantees that the assessment procedure stays in line with platform expansion and supports continual project improvement.

Overall, if users feel that the active syllabus system reduces effort and makes course information easier to follow, the project will be considered successful.

▪ ***Scenario-Based Testing***

Participants will be asked to complete everyday syllabus-related tasks, such as:

finding a due date,
checking an assessment rule,
reviewing weekly course topics.

These exercises are chosen because they represent common and realistic user actions that, in traditional syllabus designs, usually lead to misconceptions or require repeated explanation. In order to compare the two methods, these tasks will be completed on both the static syllabus and the active syllabusprototype.

During the tests, we will observe and record:

how long the tasks take,
whether users complete them successfully,
and how many errors or misunderstandings occur.

This enables us to pinpoint situations where the system fails short of user expectations and usability barriers. As new system features are added during the development process, these scenarios will be updated and expanded as necessary.

▪ ***SUS and CSUQ Questionnaires***

After the tasks, participants will fill in the:

System Usability Scale (SUS)
Computer System Usability Questionnaire (CSUQ)
These questionnaires will help us measure:

perceived usability,
usefulness,
interface quality,
and overall satisfaction.

While CSUQ will offer more in-depth insights on system usefulness, information quality, and interface design, SUS will be used to produce an overall usability score. To better reflect the final interface structure and feature set of the system, questionnaire items may be modified or added. To determine whether improvements are significant, simple statistical comparisons (e.g., between prototype versions or between formats) will be used when suitable.

The goal of these comparisons is to find usability trends and relative gains rather than flawless ratings.

▪***Comparative Analysis***

Results from the prototype will be compared to:

the traditional static syllabus, and/or
earlier design iterations,
to understand whether the new approach actually improves the user experience in practice.

This comparative approach enables us to assess the added value of the suggested system in comparison to current solutions, in addition to absolute usability levels. Variations in task performance, mistake rates, and user satisfaction will be analyzed to see whether there are still usabilityissues or opportunities for improvement.

▪*User Feedback and UX Problem Identification*

Short interviews and open-ended questions will be used to collect:

user opinions about what works well,
what feels confusing or unnecessary,
and which parts should be improved.

To find recurrent usability issues, unmet needs, and design expectations, this qualitative input will be examined using basic thematic grouping. Future design iterations will be informed by the documented UX concerns. The final design choices and suggestions for further development will be supported by these insights.

8. Ethical, Safety, and Sustainability Considerations

In planning the website, we will address a range of ethical, privacy and sustainability issues. Because students and instructors can contribute comments or feedback, clear standards and rules are essential to discourage harassment and the spread of false information. Since the platform allows interaction between students and instructors, there is a potential risk of inappropriate or misleading content. To address this, basic moderation mechanisms and clear usage guidelines will be considered in order to ensure a respectful and reliable communication environment. Data privacy is a priority as well, even if no personal data are collected initially, we will follow privacy-by-design principles (minimizing data use and securing any information) and comply with relevant student-privacy laws (e.g. FERPA in the U.S.). Finally, we will ensure long-term sustainability by adopting accessible, efficient design practices. All content will follow W3C's accessibility guidelines so that it is usable by people with disabilities in our school and the site will be built on "sustainable web design" principles (clean, efficient, open, honest, regenerative, resilient) to reduce energy use and maintenance costs over time. These measures together ensure the site remains ethical, privacy-respecting, safe, and maintainable over its usage.

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APPENDIX A

APPENDIX B

Another appendix can go here.